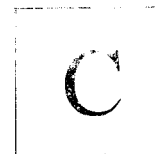


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B.Tech. Degree VI Semester Special Supplementary Examination January 2019

ME 15-1604 HEAT AND MASS TRANSFER (2015 Scheme)

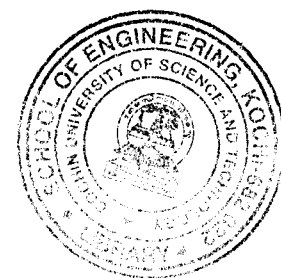
Time: 3 Hours

Maximum Marks: 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Explain Fourier law of heat conduction and the concept of thermal resistance.
- (b) What is critical thickness of insulation? Explain its significance.
- (c) Explain significance of fin effectiveness and fin efficiency.
- (d) Explain thermal boundary layer for forced convection heat transfer from flat plates.
- (e) Discuss drop wise and film wise condensations.
- (f) Explain radiation shape factor and shape factor algebra.
- (g) Explain electromagnetic radiation spectrum.
- (h) Explain Planck's law and Wien's displacement law.
- (i) How heat exchangers are classified based on the flow direction?
- (j) Define heat exchanger effectiveness and NTU.



PART B

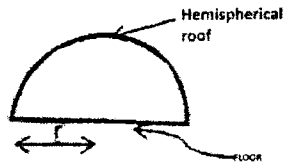
(4 × 10 = 40)

- II. (a) Derive general heat conduction equation in cylindrical coordinates. (6)
 - (b) Calculate the critical radius of insulation for asbestos [$k = 0.17 \text{ W/m}^\circ\text{C}$] surrounding a pipe and exposed to room air at 20°C with $h = 3.0 \text{ W/m}^2\text{C}$. Also determine the heat loss per unit length from a 200°C , 5 cm diameter pipe, when covered with the critical radius of insulation and without insulation. (4)
- OR**
- III. (a) Derive the expression for temperature distribution and heat transfer rate in an infinitely straight fin of rectangular profile. (6)
 - (b) A steel ball [$C_p = 0.46 \text{ kJ/kg}^\circ\text{C}$, $k = 35 \text{ W/m}^\circ\text{C}$, $\rho = 7800 \text{ kg/m}^3$], 5 cm in diameter and initially at a uniform temperature of 450°C is suddenly placed in a controlled environment in which the temperature is maintained at 100°C . The convection heat-transfer coefficient is $10 \text{ W/m}^2\text{C}$. Calculate the time required for the ball to attain a temperature of 150°C . (4)
- IV. (a) Air at 27°C and 1 atm flows over a flat plate at a speed of 2 m/s. The plate is heated over its entire length to a temperature of 60°C . (i) Calculate the hydrodynamic and thermal boundary-layer thickness at distances of 20 cm and 40 cm from the leading edge of the plate. (ii) Calculate the heat transferred from top surface in the first 20 cm of the plate. (iii) Calculate the skin friction or drag coefficient at a distance of 40 cm from leading edge of plate. (6)
 - (b) What is Reynold's analogy? Describe relation between fluid friction and heat transfer. (4)

OR

(P.T.O.)

- V. (a) A horizontal pipe 0.3 m in diameter is maintained at a temperature of 250°C in a room where the ambient air is at 15°C . Calculate the free-convection heat loss per meter of length. (5)
- (b) Explain the boiling regimes and boiling curve with the help of a suitable sketch. (5)
- VI. (a) Derive the relation for radiative heat exchange between two non-black infinite parallel planes. (6)
- (b) For a hemispherical furnace, the floor is at 700 K, and has an emissivity of 0.5. The hemispherical top is at 1000 K and has emissivity of 0.25. Determine the net radiative heat transfer per unit area from roof to floor. (4)



OR

- VII. (a) Calculate the net radiant heat exchange per m^2 area for two large parallel plates at temperatures of 727°C and 127°C respectively. The emissivities of hot and cold plates are 0.8 and 0.4 respectively. If a polished aluminium shield of emissivity 0.04 on both sides is placed between them, find the shield temperature and percentage reduction in the heat transfer. (7)
- (b) Explain the terms absorptivity, reflectivity and transmissivity. (3)
- VIII. (a) A counter flow heat exchanger is used to cool 2000 kg/hr of oil ($C_p = 2.5 \text{ kJ/kgK}$) from 105°C to 30°C by the use of water entering at 15°C . The overall heat transfer coefficient is expected to be $1.5 \text{ kW/m}^2\text{K}$. Presume that the exit temperature of the water is not to exceed 80°C . Using NTU method, make calculations for: (i) the water flow rate (ii) the surface area required for heat transfer (iii) the effectiveness of heat exchanger. (7)
- (b) What do you mean by fouling factor? What are the causes of fouling? (3)

OR

- IX. (a) Derive an expression for log mean temperature difference (LMTD) for a parallel flow heat exchanger. (5)
- (b) Steam condenses at 100°C on the external surface of the tubes of a steam condenser. The tubes are 12 in number and each is 30 mm in diameter and 10 m long. The inlet and outlet temperatures of cooling water flowing inside the tubes are 25°C and 60°C . If the water flow rate is 1.1 kg/s, calculate (i) Rate of condensation of steam (ii) The overall heat transfer coefficient based on the tube surface area. (5)
